Multi-Resolution Global Horizontal Irradiance Forecasting at Electric-Vehicle Manufacturing Locations: TimesNet and Dilated Recurrent Networks

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Abstract

This study compares TimesNet and dilated recurrent neural networks (DilatedRNN) for global horizontal irradiance (GHI) forecasting at ten locations associated with electric-vehicle factories. Using model-based estimates from the National Solar Radiation Database (NSRDB), we evaluate four tasks: hourly forecasts up to 72 hours, daily-mean forecasts up to 30 days, one-month forecasts, and exploratory forecasts up to six months. Within each task, the models share historical GHI inputs, site identifiers, forecast origins, temporal partitions, and post-processing. Performance is measured by mean absolute error averaged equally across sites (macro-MAE). Over the complete forecast horizons, TimesNet achieved lower macro-MAE at 72 hours (52.48 versus 52.96 W/m^2), while DilatedRNN led at 30 days (43.85 versus 44.08 W/m^2), one month (13.01 versus 13.89 W/m^2), and six months (14.37 versus 14.46 W/m^2). The one-month reduction relative to TimesNet was 6.38%. Daily and monthly comparisons use five-seed ensembles; the hourly comparison uses seed 42. Rankings vary across horizons and locations, and several ensemble differences are small relative to variability across seeds. The full benchmark shows that other evaluated models outperform both architectures in several tasks. Independent meteorological data provide context for selected cases after evaluation. The findings support task-specific comparisons, with descriptive conclusions limited by overlapping forecast targets and a single test year.

Keywords: global horizontal irradiance, solar forecasting, TimesNet, dilated recurrent neural network, multi-resolution forecasting, electric-vehicle factories

1. Introduction

Pathways consistent with deep emissions reductions combine a rapid expansion of low-emission electricity with the electrification of end uses [1, 2]. Declining renewable-energy costs support this transition [3], while scenario-based evidence suggests that solar photovoltaics may develop substantial self-sustaining growth before mid-century [4]. As deployment grows, the weather-dependent and diurnal variability of solar energy makes forecasts across different time horizons increasingly important for planning, scheduling, and energy management.

Industrial decarbonization requires combinations of energy efficiency, material changes, low-carbon fuels, electrification, and low-emission electricity [5, 6]. Direct electrification can reduce emissions from suitable industrial heat uses, but its potential depends on process temperature, infrastructure, and the carbon intensity of the electricity supply [7]. On-site and procured renewable electricity can therefore support manufacturing alongside other energy options [8, 9], although adoption faces site-specific financial, regulatory, technical, and organizational barriers [10].

Policy and corporate disclosures illustrate this industrial context. In the European Union, Directive (EU) 2024/1275 establishes a phased framework for solar installations on specified building categories when technically, economically, and functionally suitable [11]; it does not apply to the Brazilian, Mexican, or United States locations examined here. Nestlé reports that its KitKat production lines in Caçapava use predominantly solar electricity [12]; Mercedes-Benz describes rooftop photovoltaics and storage at Factory 56 [13]; and Tesla and Ford report solar installations at selected manufacturing sites [14, 15].

Solar forecasting serves different operational needs. Intra-hour forecasts can track cloud motion, hourly and day-ahead forecasts support short-term scheduling, and daily or monthly aggregates inform longer-term planning [16, 17]. Both temporal resolution and forecast horizon must therefore accompany terms such as short-term and long-term [18, 19]. Accuracy for a direct 72-hour sequence of hourly global horizontal irradiance (GHI) does not establish accuracy for daily means or a one-step monthly target. Likewise, comparing errors across hourly and monthly series requires accounting for differences in targets, sample sizes, and variability.

This article studies GHI at ten geographic points associated with electric-vehicle manufacturing facilities. The points span different latitudes and atmospheric regimes, providing a geographically diverse test bed. Their selection does not establish whether photovoltaic systems are installed at those facilities. The irradiance records are satellite-derived, model-based esti-

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mates from the National Solar Radiation Database (NSRDB) [20], with no factory sensor measurements or proprietary operational data. The forecasts thus characterize the solar resource at the selected grid cells, rather than factory load or photovoltaic power output.

Cabral et al. [21] evaluated very-long-term GHI forecasting at 40 climatically diverse locations using a GHI-only GPT-Neo pipeline and a broad set of competitors. Here, the focus is on comparing architectures across temporal resolutions. Building on the present authors' preliminary studies of a compact TimesNet for direct hourly forecasting and a dilated recurrent neural network (DilatedRNN) for one-step monthly forecasting, this article evaluates both architectures on matched samples within each task. The preliminary studies provide methodological background; their numerical results are excluded from the matched comparison. Cabral et al.'s numerical results are not reused.

The research question is: *how does the relative forecast accuracy of TimesNet and DilatedRNN change with temporal resolution, horizon, and location when their inputs and evaluation protocol are matched?* The article makes four contributions:

1. it defines a common multi-resolution formulation that distinguishes hourly multi-step, daily-mean multi-step, one-step monthly, and exploratory six-month forecasting;
2. it specifies matched samples, chronological partitions, forecast origins, direct outputs, post-processing, and transparent baselines for both architectures in each shared task;
3. it reports aggregate, per-location, per-horizon, and per-seed results, with the complete benchmark in the Supplementary Material documenting cases in which either architecture is outperformed by simple baselines; and
4. it selects favorable, unfavorable, geographic, and meteorological cases using reproducible rules, with climate comparisons treated as descriptive and weather conditions identified from independent meteorological data.

Section 2 reviews the forecasting literature and identifies the comparison gap. Section 3 defines GHI, temporal aggregation, forecasting horizons, TimesNet, and DilatedRNN. The following sections describe the data and feature engineering, common methodology, experimental design, and evaluation metrics, followed by the results, limitations, and conclusions.

2. Related Work

This review compares solar-forecasting studies by target, inputs, horizon, and evaluation protocol. Section 3 introduces the underlying physical and architectural concepts.

2.1. Forecasting scope and evaluation practice

Solar-forecasting reviews organize methods by forecast horizon, input source, target variable, and application [16, 17, 18, 22]. These distinctions reflect different information sets and error structures: irradiance nowcasting from sky images, hourly GHI forecasting from historical and meteorological variables, day-ahead forecasting based on numerical weather prediction, and monthly resource forecasting address different tasks. Text-mining evidence also shows that the field has diversified in both methods and applications [19]. Reviews focused on photovoltaic power similarly find that reported performance depends on the horizon, data, and optimization procedure [23]. This review therefore treats GHI, solar irradiation, and photovoltaic power as distinct targets.

Evaluation practice is equally important. Deterministic solar forecasts should be assessed against explicit baselines, with metrics selected for the decision context and reported at the relevant horizons [24]. General forecast-evaluation research also shows that percentage measures and scale-dependent errors answer different questions [25]. Voyant et al. [26] demonstrate that the appropriate naive baseline depends on temporal structure and horizon. Across multiple climates, their comparison of statistical baselines requiring no training cautions against using a single persistence rule as the sole benchmark in every setting.

2.2. From intra-hour to week-ahead forecasting

At very short horizons, cloud observations can provide information absent from a univariate irradiance history. Chow et al. [27] derived cloud maps and motion vectors from 30-s total-sky images at a distributed test bed. They evaluated cloud conditions quantitatively and the derived GHI nowcasts qualitatively. Reviews of intra-hour methods distinguish such ground-imagery approaches from satellite, numerical weather prediction, and time-series approaches [28, 29]. This literature also shows why clear-sky or annual-average performance does not establish performance during broken-cloud conditions.

For hourly time series, hybrid networks commonly combine representation learning with recurrent dynamics. Huang et al. [30] used wavelet packet decomposition, convolution, long short-term memory (LSTM), and multilayer perceptron branches for 1-h-ahead prediction at three United States sites. Their inputs included irradiance history, meteorological variables, and next-hour forecasts of temperature, relative humidity, and wind speed. Guariso et al. [31] compared feed-forward and LSTM networks using recursive, horizon-specific, and multi-output strategies for horizons of up to 12 h. Their evaluation included persistence and clear-sky baselines and an analysis of transfer across sites. These studies differ in both model complexity and how they generate future steps.

Longer horizons can also retain hourly outputs. Cheng et al. [32] combined one-dimensional convolution and LSTM components for day-ahead to week-ahead daytime irradiance prediction, using recent irradiance and a small set of weather variables. Zhao et al. [33] coupled improved complete ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) to TimesNet, combining the decomposed solar-radiation sequences with meteorological data for week-ahead hourly forecasting. This application establishes the relevance of TimesNet to solar forecasting, but its decomposition stage and week-ahead protocol do not isolate the effect of the TimesNet backbone or provide a matched comparison with DilatedRNN.

2.3. Coarser resolution, multiple locations, and climate context

Coarser targets lead to a different learning problem. Ozoegwu [34] forecast monthly-mean daily global solar radiation with nonlinear autoregressive, exogenous-input, and hybrid neural formulations. Jung et al. [35] examined recurrent modeling for long-term photovoltaic-power forecasting, whose target also depends on the conversion system. Cabral et al. [21] evaluated a GHI-only GPT-Neo pipeline and contemporary competitors over 40 locations and a very-long-term horizon. That work provides a precedent for evaluating long-horizon forecasts across locations; the present study focuses on comparing two time-series architectures with matched temporal resolution and forecast design.

Cross-location learning can increase the effective sample size by sharing parameters. It also changes the modeling setup from separate local models to a shared forecasting model [36]. The broader recurrent neural network (RNN) literature emphasizes the need to distinguish architectural capacity from window construction, optimization, and evaluation design [37]. For seasonal and climate context, Zhu et al. [38] evaluated several deep time-series models across seasons and forecast horizons ranging from minutes to 24 h, using different input combinations and training-set lengths. Such designs describe performance variation across conditions, but causal attribution requires additional support from the study design.

The Supplementary Material summarizes these representative primary studies in three groups organized by forecast horizon and geographic or climate context. The tables compare dimensions relevant to this article's research question, without ranking reported errors across studies with different targets, sites, horizons, inputs, and test protocols.

2.4. Research gap and positioning

The reviewed studies provide evidence for specific architectures and horizons, but do not establish how TimesNet and DilatedRNN compare under a common protocol across temporal resolutions. Zhao et al.'s TimesNet application predicts week-ahead hourly solar radiation after signal decomposition [33]; monthly neural studies use different targets and architectures [34]; and the very-long-term cross-location study centers on a language-model backbone [21]. Ranking their published errors would therefore confound model choice with temporal aggregation, information availability, location set, and evaluation protocol.

The present study addresses this comparison through a common experimental design. At each resolution, TimesNet and DilatedRNN share the same samples, chronological forecast origins, admissible inputs, post-processing rules, baselines, and aggregation across locations. Evaluation distinguishes hourly multi-day forecasts from daily aggregates and treats the primary one-step monthly task separately from the exploratory six-month task. These choices address the comparability limitations identified in solar-forecast verification and benchmarking research [24, 26].

3. Theoretical Background

Following the empirical review in Section 2, this section defines GHI, temporal aggregation, and forecast design before describing the mechanisms of TimesNet and DilatedRNN.

3.1. Global horizontal irradiance and temporal support

Global horizontal irradiance (GHI) is the instantaneous solar radiant flux received per unit area by a horizontal surface. When the Sun is above the horizon,

$$\mathrm{GHI} = \mathrm{DHI} + \mathrm{DNI}\cos(\theta_z), \tag{1}$$

where DHI is diffuse horizontal irradiance, DNI is direct normal irradiance, and θ_z is the solar zenith angle [39]. Irradiance is a power density, conventionally expressed in W/m^2. Integrating irradiance over time yields irradiation, which is an energy density.

The NSRDB product used in this study estimates the radiative components from satellite observations and physical models [20]. The study uses the model-based estimate at the grid cell nearest each factory coordinate. Converting GHI into photovoltaic power would additionally require information about module area, orientation, efficiency, temperature, losses, and plant operation.

Let $y^{(1\mathrm{h})}_{\ell,t}$ be hourly mean GHI for location ℓ. The mean irradiance at a coarser temporal support Δ is

$$y^{(\Delta)}_{\ell,\tau} = \frac{1}{N_{\ell,\tau}} \sum_{t \in \mathcal{I}^{(\Delta)}_{\ell,\tau}} y^{(1\mathrm{h})}_{\ell,t}, \tag{2}$$

where $\mathcal{I}^{(\Delta)}_{\ell,\tau}$ contains the valid hourly samples in interval τ and $N_{\ell,\tau}$ is their number. Equation (2) preserves the units of W/m^2 and defines a daily or monthly mean, rather than accumulated irradiation. Coverage requirements must be applied before aggregation so that intervals with missing hours are not treated as complete.

3.2. Forecast resolution, context, and horizon

For a temporal resolution Δ, let $\mathbf{X}_{\ell,t} \in \mathbb{R}^{C\times P}$ contain C observations of P admissible variables, all observed strictly before target interval t. Throughout the article, the origin label t identifies the first predicted interval; the GHI value for that interval is unavailable when the forecast is issued. A deterministic multi-output forecaster is a mapping

$$\widehat{\mathbf{y}}_{\ell,t:t+H-1} = f_\theta(\mathbf{X}_{\ell,t}, \mathbf{z}_\ell) \in \mathbb{R}^H, \tag{3}$$

where $\mathbf{z}_\ell$ denotes static location information when used, C is the context length, and H is the number of predicted intervals. For fixed-duration hourly and daily intervals, the corresponding spans are $C\Delta$ and $H\Delta$; monthly spans follow calendar months of unequal length. Thus, 72 hourly outputs and three daily outputs have the same nominal duration but different targets and information content. Likewise, predicting one monthly mean is a one-step monthly forecast, not a multi-month trajectory [16, 19].

In a direct multi-output strategy, all H values in (3) are produced jointly. A recursive strategy instead feeds one-step predictions back as inputs and may accumulate error as the horizon increases [31]. The forecasting strategy must therefore be held constant when comparing models. A local model estimates separate parameters for each location; a global model shares θ across locations and may condition on $\mathbf{z}_\ell$ [36]. Sharing can increase the effective training sample, but it does not guarantee equal local performance.

Persistence, diurnal or seasonal naive forecasts, and simple autoregressive models provide baselines for assessing gains beyond continuity and seasonality. Their suitability depends on resolution and horizon, so evaluation may require more than one baseline [26].

3.3. TimesNet

TimesNet models one-dimensional sequences through a set of period-conditioned two-dimensional representations [40]. Let $\mathbf{X} \in \mathbb{R}^{B\times T\times C_x}$ be a batch processed by a TimesBlock, and let $F_{b,f,c}$ denote the temporal fast Fourier transform (FFT) of sample b at frequency bin f and channel c. The compact variant evaluated here selects periods independently for each sample, averaging amplitudes only over that sample's channels,

$$A_{b,f} = \frac{1}{C_x}\sum_{c=1}^{C_x}\left|F_{b,f,c}\right|. \tag{4}$$

After excluding the zero-frequency component, each sample uses its k largest amplitudes to select frequency indices $f_{b,i}$ and integer periods

$$\{f_{b,1},\dots,f_{b,k}\} = \underset{1\le f\le\lfloor T/2\rfloor}{\arg\,\text{top-k}}\, A_{b,f}, \qquad p_{b,i} = \left\lfloor \frac{T}{f_{b,i}} \right\rfloor. \tag{5}$$

The selected amplitudes, $a_{b,i} = A_{b,f_{b,i}}$, determine the aggregation weights.

For each $p_{b,i}$, the sequence is zero-padded to $L_{b,i} = p_{b,i}\lceil T/p_{b,i}\rceil$ and reshaped so that one axis indexes position within a period and the other indexes successive periods. A two-dimensional Inception-style convolution then learns intraperiod and interperiod variations. If $\mathcal{G}_{b,i}$ is the processed tensor and $\mathcal{R}_{b,i}^{-1}$ restores chronological order, the period-specific outputs are combined using weights derived from each sample's amplitudes

$$\begin{aligned}\alpha_{b,i} &= \frac{\exp(a_{b,i})}{\sum_{j=1}^{k}\exp(a_{b,j})},\\ \widetilde{\mathbf{X}}_b &= \mathbf{X}_b + \sum_{i=1}^{k}\alpha_{b,i}\left[\mathcal{R}_{b,i}^{-1}(\mathcal{G}_{b,i})\right]_{1:T}.\end{aligned} \tag{6}$$

The residual connection preserves the original representation. Before value embedding, the implementation standardizes each input window using its own temporal mean and population standard deviation, with 10^{-5} added to the variance. It reverses this transformation at the output. This causal window standardization follows the external site-wise min–max transformation described in Section 4.1. Stacked TimesBlocks and a task-specific projection produce the forecast. Because period selection and aggregation weights are computed for each sample independently, changing other windows in the batch does not change the information used for that sample's forecast. TimesNet has also been combined with signal decomposition for week-ahead hourly solar forecasting [33], although decomposition is not intrinsic to the architecture.

3.4. Dilated recurrent neural networks

A DilatedRNN changes the recurrent connection pattern rather than reshaping the sequence [41]. Let $\mathbf{c}_t^{(q)}$ denote the recurrent state in layer q, and let $\mathbf{x}_t^{(q)}$ be the layer input, with $\mathbf{x}_t^{(1)} = \mathbf{x}_t$ and $\mathbf{x}_t^{(q)} = \mathbf{c}_t^{(q-1)}$ for $q > 1$. A dilated recurrent layer updates

$$\mathbf{c}_t^{(q)} = \phi_q\left(\mathbf{x}_t^{(q)}, \mathbf{c}_{t-d_q}^{(q)}\right), \tag{7}$$

where ϕ_q can be an ordinary RNN, gated recurrent unit, or LSTM cell [41, 42]. In that layer, the recurrent connection from $t-1$ is replaced by one from $t-d_q$; dilation does not add a residual connection.

With an integer base $M > 1$, the standard exponentially increasing schedule is

$$d_q = M^{q-1}, \qquad q = 1,\dots,Q. \tag{8}$$

The first layer retains local recurrence when $d_1 = 1$, while higher layers connect states at progressively coarser temporal separations. This schedule shortens recurrent paths between distant observations and represents dependencies at multiple temporal scales. The interleaved recurrent chains are recombined at their original time indices, preserving the intervening observations without downsampling. An output head maps the resulting states to one or several forecast steps.

TimesNet and DilatedRNN encode multi-scale dependence through different inductive biases: TimesNet discovers dominant Fourier periods and applies two-dimensional convolutions, while DilatedRNN uses a fixed hierarchy of sparse recurrent lags. Comparing them requires a matched context, forecasting strategy, horizon, and evaluation protocol. Differences in capacity and response to optimization still limit attribution of performance to individual architectural mechanisms.

3.5. Atmospheric variability and climate context

Solar geometry creates a deterministic diurnal envelope, whereas clouds and other atmospheric constituents modulate the irradiance reaching the surface. Cloud advection can create rapid GHI changes and photovoltaic power ramps [27, 43]. Aggregation reduces some high-frequency variation but does not remove persistent cloudy regimes, synoptic variability, or seasonality.

Low GHI alone does not establish precipitation because it can also reflect solar elevation, cloud optical properties, aerosols, or the aggregation window. Describing conditions as rainy or cloudy therefore requires independent meteorological data aligned in time and location. Reanalysis products can provide this context, subject to their spatial resolution and model uncertainty [44, 45].

Köppen–Geiger classes summarize long-term temperature and precipitation regimes [46]. They are useful for describing the geographic diversity of the sampled locations, but they are not dynamic forecast inputs unless explicitly encoded. With ten sites, differences between classes are exploratory associations rather than causal estimates of climate effects on forecast error.

4. Unified Proposed Approach

4.1. Data and Feature Engineering

4.1.1. Forecasting data and spatial sampling

The forecasting target is mean global horizontal irradiance (GHI), expressed in W/m^2 at each temporal resolution. The data come from the GOES Aggregated Physical Solar Model v4, distributed through the National Solar Radiation Database (NSRDB) [20, 47]. This satellite-informed physical model provides gridded irradiance estimates. Factory coordinates select the associated NSRDB grid cells; throughout the article, *location* denotes one of these grid points. The records are not on-site sensor measurements.

The ten locations form a common panel covering Brazil, Mexico, and the United States, but the temporal tasks use two independently acquired NSRDB archives. The official hourly archive contains native 30-min records that its pipeline aggregates to one-hour means. The daily archive was requested separately at a 60-min interval and exported as arithmetic civil-day means; monthly GHI is then computed as the arithmetic mean of those daily values in each civil month. Both archives cover the same ten factory-associated locations and the nominal 2019–2024 study period, but the daily/monthly records are not reaggregations of the exact HDF5 records used by the hourly experiment. Retrieval resolution and product revisions may therefore introduce differences between the archives, further motivating error comparisons within each task. Observations from 2019 provide the initial context needed by tasks whose first forecast origin occurs in 2020.

Table 1 distinguishes model inputs from indexing and contextual fields. TimesNet, DilatedRNN, LSTM, and XGBoost receive the same historical GHI sequence and site identifier. Calendar variables, coordinates, climate classes, and meteorological descriptors are excluded from their inputs. The deterministic seasonal and climatological baselines additionally use target calendar labels to retrieve historical values or fitted means, as specified in the Supplementary Material.

4.1.2. Data quality and leakage prevention

Each stored source is audited according to its schema before modeling. Timestamp and GHI fields are parsed, the records are sorted chronologically, and the resulting index must be unique, monotonic, regular, and common across locations. Invalid timestamps, nonnumeric or non-finite GHI, negative GHI, duplicates, discontinuous grids, and an inconsistent stored site identity are rejected. Unsorted records are reordered before these checks. Product and retrieval metadata are retained as provenance rather than supplied to the forecast models. Monthly aggregation follows the daily checks.

Table 1: Fields used by the forecasting and contextual pipelines.

Role	Field	Use
Target and lagged input	GHI	Direct forecast target and causal history at the task resolution.
Static model input	Site identifier	Embedding for neural models and one-hot encoding for XGBoost.
Temporal index	Timestamp	Ordering, partition boundaries, forecast origins, and target alignment; not an additional covariate.
Spatial index	Factory latitude/longitude	Selection of the NSRDB grid cell and sampling of contextual maps; not a learned input.
Learned transform	Site-wise min–max scale	Fitted strictly before the validation or test cutoff and reused without clipping.
Hourly physical rule	Solar elevation	Sets post-processed nighttime forecasts to zero; not an input to model fitting.
Context only	Köppen–Geiger class and NASA POWER fields	Post-hoc geographic and adverse-weather interpretation; never used for training, selection, or forecasting.

For a site ℓ, the matched input at origin t is

$$\mathbf{x}_{\ell,t} = (y_{\ell,t-c}, \ldots, y_{\ell,t-1};\, \ell)\,, \tag{9}$$

where c is the task-specific context length (C in Eq. (3)) and ℓ is the fixed site identifier. Every target vector starts at t and must remain within its assigned partition. Each input window contains only observations preceding its forecast origin, while normalization parameters use only observations before the corresponding fitting cutoff. Hourly and daily labels identify the beginning of the target interval. Monthly records use month-end labels to identify the civil month being forecast; these labels are not physical issuance times. The monthly information set contains only the preceding completed months.

Min–max parameters are estimated separately for each site. Model selection uses scales fitted to observations strictly before 1 January 2023. After the epoch count has been selected, the refitted model uses new scales fitted strictly before 1 January 2024. The same parameters transform the corresponding inputs and targets and invert the forecasts. Values outside the training range are not clipped, so extrapolation beyond that range remains visible.

4.1.3. Geographic and climatic context

Table 2 reports the canonical factory coordinates, the map identifier used in Figure 1, and the 1980–2016 Köppen–Geiger class sampled at those coordinates. Classes and their confidence values come from the aligned 0.008333-degree (nominal 1-km) products of Beck et al. [46]. Sampling uses the raster cell containing each WGS84 point, without interpolation. The 0.5-degree product is used only as the background of the Americas map and never to assign a site's class.

Class confidence is the percentage of Beck et al.'s 12 ensemble maps that assign the modal class to a cell [46]. Nine sites have a class confidence of 100%. BMW San Luis Potosí is

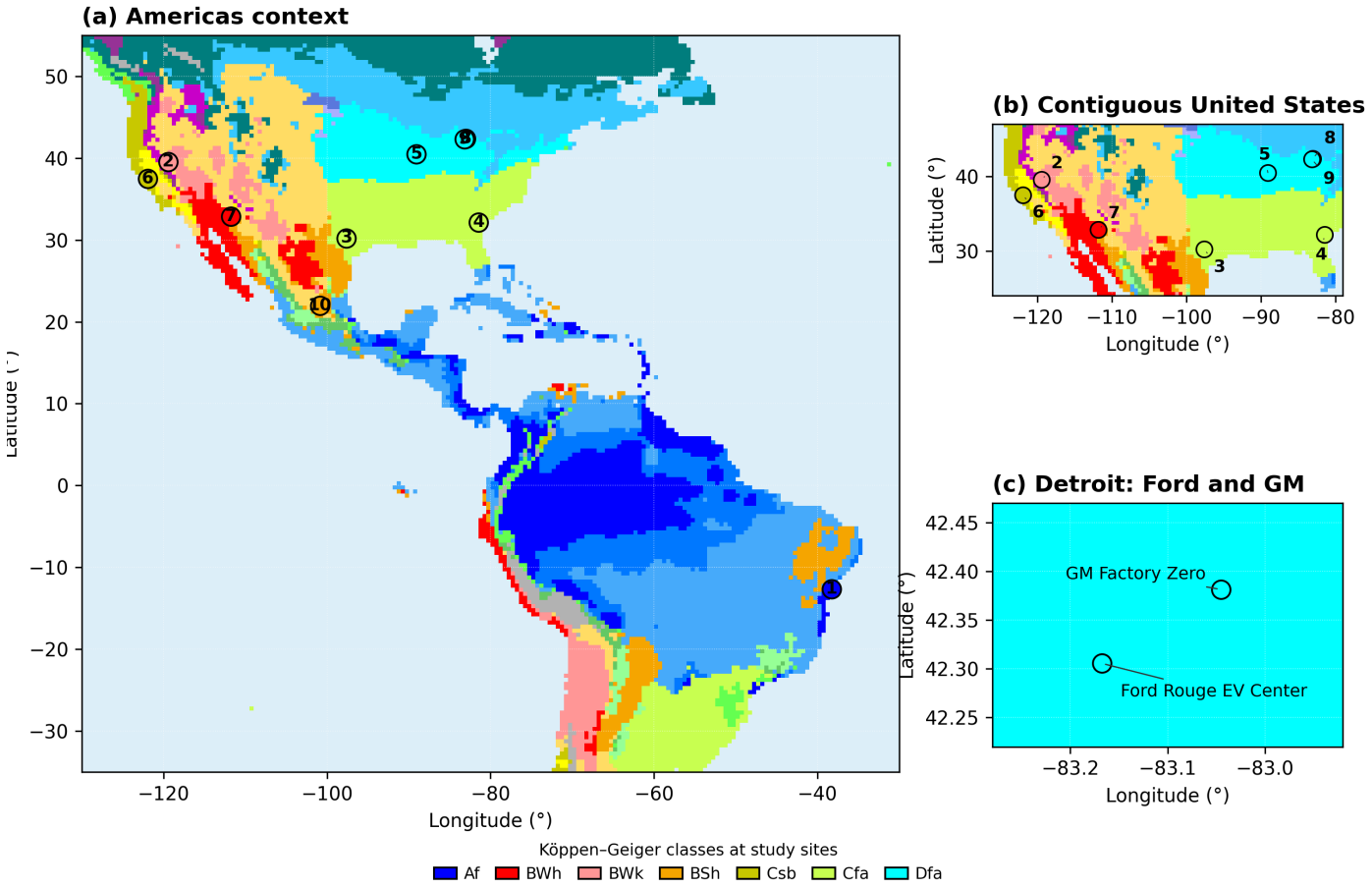


Figure 1: Geographic and climatic context of the ten factory-associated locations. The Americas panel uses the present 0.5-degree Köppen–Geiger raster for visual context; site assignments and confidence values are sampled from the present nominal 1-km raster. Numbered markers match the Map ID column in Table 2. The United States and Detroit insets separate spatially close labels. The map is descriptive and its climate fields are not forecast-model inputs. Source: Beck et al. [46].

the exception: its BSh assignment has 50% confidence and is therefore treated as uncertain in interpretation. The Supplementary Material documents the climate codes, confidence values, raster-cell indices, source metadata, and file hashes.

4.1.4. Independent post-hoc meteorological context

Daily NASA POWER fields for 2024 provide independent context for interpreting forecast errors [48]. These gridded estimates describe weather around each location. The variables are corrected precipitation, cloud amount, all-sky surface shortwave irradiation, and clear-sky surface shortwave irradiation in local solar time. The complete table contains 366 dates for each of the ten locations.

The study defines three descriptive tags independently of forecast error: precipitation-indicated when corrected precipitation is at least 1 mm/day, high-cloud when cloud amount is at least 80%, and adverse when either condition holds. These study-specific thresholds are applied after forecasts and errors have been calculated. NASA POWER values are excluded from model inputs, training, validation, refitting, and model or metric selection. POWER dates in local solar time are matched to the NSRDB calendar with fixed local offsets, using factory coordinates rather than exact NSRDB grid-cell centers. The resulting match provides approximate daily context, without confirming weather at individual forecast hours or establishing the causes of forecast errors.

4.2. Methodology

The method compares TimesNet [40] and DilatedRNN [41] on identical windows and forecast origins within each temporal task. Both models perform continuous GHI regression; the quantized forecasts and errors from the preliminary DilatedRNN study are excluded. Figure 2 summarizes the three stages. Algorithm 1 specifies temporal cutoffs, epoch selection, refitting, and the hourly extension. Algorithms 2 and 3 describe the implemented forward passes. The Supplementary Material provides additional data checks and evaluation procedures.

4.2.1. Block A: data preparation and temporal windows

Block A checks the source series, constructs the required temporal aggregates, assigns a stable site identifier, and creates chronological partitions. It then fits site-wise min–max scales using observations before 2023 for model selection and before 2024 for the final refit. The input is the historical GHI window and site identifier defined in Eq. (9). Keeping this information

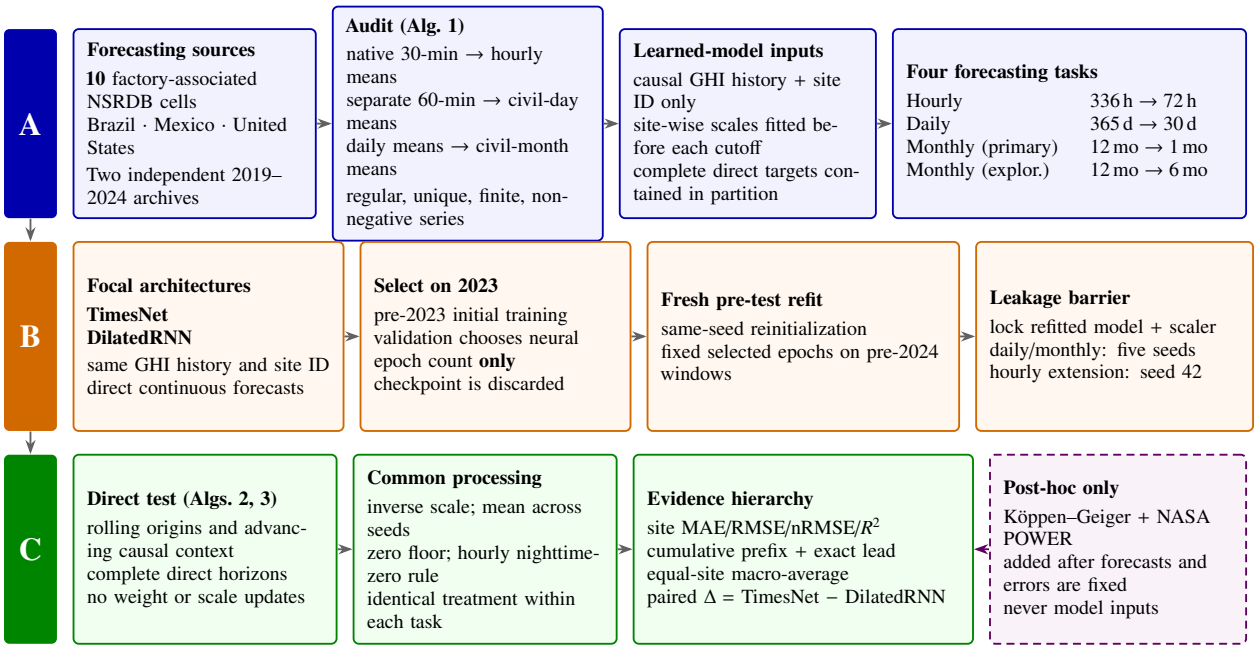


Figure 2: End-to-end matched forecasting design for TimesNet and DilatedRNN. (A) Independent NSRDB archives are audited and converted into four direct-output tasks; learned models receive causal GHI and site identity. (B) Model selection and fresh pre-test refitting follow Algorithm 1. (C) Direct test forecasts from Algorithms 2 and 3 receive common physical post-processing before location-first aggregation and paired comparison; Köppen–Geiger and NASA POWER fields enter only after errors are fixed.

Table 2: Factory reference coordinates, map identifiers, and present-day 1-km Köppen–Geiger assignments.

Map ID	Factory location	Country	Lat., lon. (°)	Class (conf.)
1	BYD Camaçari	Brazil	−12.6734, −38.2812	Af (100%)
2	Tesla Gigafactory Nevada	USA	39.5404, −119.4391	BWk (100%)
3	Tesla Gigafactory Texas	USA	30.2219, −97.6188	Cfa (100%)
4	Hyundai Metaplant Georgia	USA	32.1605, −81.4510	Cfa (100%)
5	Rivian Normal	USA	40.5093, −89.0545	Dfa (100%)
6	Tesla Fremont Factory	USA	37.4927, −121.9415	Csb (100%)
7	Lucid AMP 1 Casa Grande	USA	32.8569, −111.7784	BWh (100%)
8	GM Factory Zero	USA	42.3815, −83.0453	Dfa (100%)
9	Ford Rouge EV Center	USA	42.3056, −83.1676	Dfa (100%)
10	BMW San Luis Potosí	Mexico	21.9683, −100.8492	BSh (50%)

Class descriptions from the distributed Beck et al. legend: Af, tropical rainforest; BWh/BWk, hot/cold arid desert; BSh, hot arid steppe; Cfa, temperate, no dry season, hot summer; Csb, temperate, dry summer, warm summer; and Dfa, cold, no dry season, hot summer. Coordinates are factory reference points; they select the associated NSRDB cell and are not sensor coordinates.

set constant gives both architectures access to the same predictors.

4.2.2. *Block B: matched development of TimesNet and DilatedRNN*

The focal comparison evaluates TimesNet [40] and a direct DilatedRNN [41]. Both architectures receive the same scaled GHI sequence and site identifier, encoded through a learned embedding. TimesNet selects its Fourier periods separately for every input window; other windows in an inference batch cannot alter its periods or forecast. Each architecture–seed instance is fitted globally across sites, sharing parameters while retaining site identity [36]. Both models produce the complete target vector in one forward pass, without returning predicted GHI recursively as input.

For each architecture–seed pair, the 2023 validation windows select only the number of training epochs. The validation checkpoint is then discarded. The final instance is trained from a fresh initialization under the same seed on all declared pre-test windows for exactly the selected number of epochs. The hourly extension reuses the existing official TimesNet forecasts and adds only DilatedRNN with seed 42. The two forecasts are aligned by site, origin, target timestamp, and lead.

Algorithm 1 describes this two-architecture protocol. The supplementary benchmark reports the additional reference models, their configurations, and the resolution-specific definitions of persistence, annual seasonal naive, and calendar climatology.

Evaluation follows a rolling-origin protocol. At each eligible origin, the input window advances to include the latest historical observations, including earlier observations from 2024 during the test period. Model parameters and the refit scale remain fixed throughout that year. This retrospective walk-forward design evaluates successive forecasts using only the information available before each target interval.

Algorithm 1 Matched TimesNet–DilatedRNN preparation and refit

Require: Task (r, C, H); site archives; fixed cutoffs and settings
Ensure: Origin-aligned TimesNet and DilatedRNN forecasts and metrics
Load the archive for r; audit the common time grid and GHI
For monthly tasks, average audited daily values by civil month
Build initial training (adjustment) windows $\mathcal{W}_A$ and validation $\mathcal{W}_V$, refit $\mathcal{W}_R$, and test $\mathcal{W}_T$ windows with context $y_{t-C:t-1}$, targets $y_{t:t+H-1}$, and site ID
Reject target windows crossing a cutoff; verify origin counts
Fit site scales S_A before 2023 and S_R before 2024
if hourly extension **then**
    Verify hashes; load the official TimesNet seed-42 forecasts
    $\mathcal{M} \leftarrow \{\text{DilatedRNN}\}$; $\mathcal{S} \leftarrow \{42\}$
else
    $\mathcal{M} \leftarrow \{\text{TimesNet}, \text{DilatedRNN}\}$
    $\mathcal{S} \leftarrow \{11, 23, 42, 67, 89\}$
end if
for each $(m, s) \in \mathcal{M} \times \mathcal{S}$ **do**
    Initialize with s; fit $S_A(\mathcal{W}_A)$ using Adam
    Select $E^\star$ by early stopping on $S_A(\mathcal{W}_V)$ MSE
    Retain the validation forecasts from epoch $E^\star$
    Discard weights; reinitialize with the same seed s
    Fit $S_R(\mathcal{W}_R)$ for exactly $E^\star$ epochs
    Forecast every $S_R(\mathcal{W}_T)$ context with fixed weights (Algorithms 2 and 3)
    Invert S_A for validation and S_R for test forecasts
    Retain the unbounded seed forecasts in W/m^2
end for
For daily/monthly tasks, average raw forecasts across seeds
Post-process each new seed and ensemble forecast: apply the zero floor and, if hourly, the nighttime mask
Verify common $(\ell, t, \text{lead}, \text{target})$ keys
Compute site metrics, then macro summaries with equal site weights

4.2.3. Direct architecture implementations

Algorithm 2 specifies the compact TimesNet forward pass used in the comparison. The site-scaled context is standardized within the current window, embedded with position and site identity, and processed using periods selected separately for that sample. After the TimesBlock, a direct temporal projection maps the context to the complete forecast horizon. Sample-wise period selection, the site embedding, and this post-block temporal head are design choices of the evaluated compact adaptation of TimesNet [40]. The reported results characterize that implementation under the matched protocol. The final step reverses the within-window standardization, returning predictions in site-normalized units. Block C then converts predictions to physical units and applies common post-processing.

Algorithm 2 Direct forward pass of the compact TimesNet

Require: Fitted TimesNet of width D; site-scaled $\mathbf{x} \in \mathbb{R}^{C\times 1}$; site ℓ
Ensure: H predictions on the site-specific min–max scale
$\mu \leftarrow \text{mean}(\mathbf{x})$; $\sigma \leftarrow \sqrt{\text{var}_{\text{pop}}(\mathbf{x}) + 10^{-5}}$
$\mathbf{u} \leftarrow (\mathbf{x} - \mu)/\sigma$
$\mathbf{Z} \leftarrow \text{ValueEmbed}(\mathbf{u}) + \text{PositionEmbed}$
$\mathbf{Z} \leftarrow \mathbf{Z} + \text{SiteEmbed}(\ell)$
Apply embedding dropout during training only
for each TimesBlock **do**
    $a_f \leftarrow D^{-1} \sum_{q=1}^{D} |\text{rFFT}_t(\mathbf{Z})_{f,q}|$
    Select the top-k indices $f_i \in \{1, \ldots, \lfloor C/2 \rfloor\}$ by a_f
    for each selected f_i **do**
        $p_i \leftarrow \lfloor C/f_i \rfloor$; $L_i \leftarrow \lceil C/p_i \rceil p_i$
        Zero-pad to L_i steps; fold into L_i/p_i cycles
        Apply two Inception banks with GELU between them (mean of the 1×1 and 3×3 convolution outputs)
        Unfold and trim to C steps, obtaining $\mathbf{U}_i$
    end for
    $\alpha_i \leftarrow \exp(a_{f_i}) / \sum_{j=1}^{k} \exp(a_{f_j})$
    $\mathbf{Z} \leftarrow \text{LayerNorm}(\mathbf{Z} + \sum_i \alpha_i \mathbf{U}_i)$
end for
$\mathbf{F} \leftarrow \text{TemporalProject}_{C\to H}(\mathbf{Z}) \in \mathbb{R}^{H\times D}$
$\widehat{\mathbf{u}} \leftarrow \text{ChannelProject}(\mathbf{F})$
return $\sigma\widehat{\mathbf{u}} + \mu$

Algorithm 3 specifies the alternative recurrent path. Each layer shares its tanh-RNN weights across interleaved chains with the same dilation, then reassembles the outputs in temporal order. The last state and the site embedding feed a direct output head. The explicit forward passes make clear which operations differ while both architectures retain the same context, site identity, target vector, and subsequent evaluation.

Algorithm 3 Direct forward pass of the DilatedRNN

Require: Fitted DilatedRNN; site-scaled $\mathbf{x} \in \mathbb{R}^{C\times 1}$; site ℓ
Ensure: H predictions on the site-specific min–max scale
$\mathbf{Z} \leftarrow \mathbf{x}$
for each dilation $d \in (1, 2, 4)$ **do**
    Use this layer's shared 16-unit tanh-RNN
    Process $\mathbf{Z}_{j, j+d, \ldots}$ from zero states, $0 \le j < d$
    $\mathbf{Z} \leftarrow$ outputs in time order, of shape $C \times 16$
end for
$\mathbf{v} \leftarrow [\mathbf{Z}_{C-1}; \text{SiteEmbed}(\ell)] \in \mathbb{R}^{24}$
$\mathbf{a} \leftarrow \text{ReLU}(\text{Linear}_{24\to 16}(\mathbf{v}))$
return $\text{Linear}_{16\to H}(\mathbf{a})$

4.2.4. Block C: physical consistency and evaluation

For daily and monthly learned models, the ensemble is the arithmetic mean of the five inverse-scaled forecasts before physical post-processing. If $\widehat{y}^{(s)}$ is the unbounded forecast in W/m^2 from seed s, the reported ensemble forecast is

$$\widehat{y}^{\text{ens}} = \max\left(0, \frac{1}{5}\sum_{s=1}^{5} \widehat{y}^{(s)}\right). \tag{10}$$

Seed-level forecasts receive their own zero floor before their metrics are computed. Averaging forecasts and averaging their errors are distinct operations.

All forecasts receive the same non-negativity floor after inverse scaling. Daily and monthly forecasts have no upper truncation. Hourly forecasts also use solar elevation computed at the center of each hourly interval; predictions are set to zero

when elevation is nonpositive, following the deterministic geometry implemented with `pvlib` [49]. Solar elevation constrains the output without entering model fitting.

The location is the unit of the descriptive comparison. For each metric and horizon scope, we report the TimesNet–DilatedRNN difference, its mean and median across the ten sites, and the number of sites won by TimesNet. The treatment of overlapping origins is defined in Section 4.4. NASA POWER data are added after evaluation, using the 1 mm/day and 80% thresholds in Section 4.1 to contextualize selected cases.

4.3. Experimental Setup

The four forecasting tasks appear in Block A of Figure 2. The Supplementary Material summarizes their context lengths, outputs, evaluation horizons, and origin counts. Within each task, all models share sites, origins, targets, fitting cutoffs, and post-processing. MAE and RMSE are reported in W/m^2 at every resolution, but they describe different temporal aggregates and are interpreted separately across tasks.

4.3.1. Common matched protocol

The daily and monthly experiments contain three deterministic references (persistence, annual seasonal naïve, and calendar climatology) and four learned models (XGBoost, LSTM, TimesNet, and DilatedRNN). Every learned model receives only causal GHI and site identity. Neural models use the five seeds 11, 23, 42, 67, and 89, and XGBoost is likewise refitted under each of those seeds because row and column subsampling are stochastic. For each learned model, we evaluate both individual-seed forecasts and their arithmetic-mean ensemble, applying the post-processing in Section 4.2. Deterministic baselines are evaluated once. The primary comparison uses macro-MAE, calculated by averaging site-level MAEs with equal weights. Complementary results include RMSE, nRMSE, and R^2 at cumulative horizons and exact leads.

For all neural models, the selection and fresh-refit procedure in Algorithm 1 uses early stopping on 2023 site-normalized validation MSE. Initial training uses the adjustment partition recorded in the saved artifacts; the selected epoch count governs the same-seed refit on pre-2024 windows. This selection objective differs from the primary test ranking by macro-MAE in physical units. XGBoost uses separate initial-training and refit instances with a fixed tree budget. Only refitted instances generate test forecasts.

Block B of Figure 2 summarizes this selection and refitting sequence.

Forecasts follow the rolling-origin protocol in Section 4.2. Hourly forecasts are issued once per day at local midnight, daily forecasts advance one origin per day, and monthly forecasts advance one origin per civil month.

4.3.2. Hourly protocol and DilatedRNN extension

The official hourly task uses 336 past hours (14 days) and produces the next 72 hours directly. The reported checkpoints are 24, 48, and 72 hours; they are prefixes and exact leads extracted from the same 72-hour output, not separately trained models. Initial training origins run from 15 January 2019 through 29 December 2022 (1,445 per site), and validation origins run from 1 January through 29 December 2023 (363 per site). The pre-test refit combines origins from 15 January 2019 through 29 December 2023 (1,810 per site). The retrospective test contains 363 origins from 1 January through 28 December 2024. The last 72-hour window ends at 23:00 on 30 December in the fixed local-time index; corresponding UTC timestamps can extend into 31 December for western locations.

The official artifact fixes seed 42 and contains persistence, annual seasonal naïve, XGBoost, LSTM, and TimesNet predictions. The extension trains only a direct DilatedRNN with seed 42 using the same 336-hour context and 72-hour output. Its forecasts are matched one-to-one with the existing predictions by site, origin, target timestamp, and lead. Target equality and input hashes are checked to ensure that the extension preserves the original data and comparator forecasts.

4.3.3. Daily protocol

The daily primary task uses 365 daily means to predict a 30-day vector. Metrics are evaluated at 1, 3, 7, 14, and 30 days. Initial training origins extend from 1 January 2020 through 2 December 2022, with targets ending no later than 31 December 2022 (1,067 origins per site). Validation uses 1 January–2 December 2023 (336 origins), and refit uses 1 January 2020–2 December 2023 (1,432 origins). The 2024 test uses 1 January–2 December and contains 337 complete origins per site because 2024 is a leap year.

4.3.4. Monthly primary and exploratory protocols

Civil-month means are calculated from the audited daily series. The primary monthly task uses 12 months to predict the next month. It has 36 initial training origins in 2020–2022, 12 validation origins in 2023, 48 refit origins in 2020–2023, and 12 test origins in 2024, for every site.

The exploratory monthly task uses the same 12-month context and directly predicts six months. Its declared checkpoints are months 1, 3, and 6. The final origin in each partition must fall no later than July so that all six targets remain inside its final year. Initial training therefore contains 31 consecutive monthly origins from January 2020 through July 2022, validation contains seven from January through July 2023, refit contains 43 from January 2020 through July 2023, and test contains seven from January through July 2024. Earlier complete years in initial training and refit retain all twelve origins. This task is exploratory because each site contributes only seven test origins, each with a six-month output.

For the six-month task, exact leads are evaluated on different target-month sets: lead 1 covers January–July, whereas lead 6 covers June–December. Its horizon profile therefore combines lead-time and seasonal effects and is not interpreted as an isolated degradation curve.

4.3.5. Fixed optimization settings

Table 3 gives the compact configuration needed to interpret the main comparison. The Supplementary Material records the

Table 3: Compact fixed configuration of the matched implementation.

Component	Fixed setting
Matched input	Causal GHI sequence plus fixed site identifier only
Regression/output	Continuous GHI; direct vector of task-specific length
Neural optimization	Adam; MSE loss; learning rate 10^{-3}; weight decay 10^{-5}
Batch size	128 for hourly/daily; 32 for monthly
Epoch selection	Hourly/daily: maximum 30, patience 5; monthly: maximum 300, patience 30
Seeds	Daily/monthly: 11, 23, 42, 67, 89; official hourly and DilatedRNN extension: 42
TimesNet	$d_{\text{model}} = 8$, $d_{\text{ff}} = 16$, one TimesBlock, top-$k = 3$, two kernels, dropout 0.1
DilatedRNN	Dilations $(1, 2, 4)$; 16 units/layer; dense width 16; dropout 0
LSTM	One layer; hidden width 32; direct output head
Site embedding	Width 8 for all neural models
XGBoost	150 trees; depth 3; learning rate 0.05; row and column subsampling 0.8; histogram tree method
Post-processing	Zero floor for all tasks; hourly values also set to zero when solar elevation is nonpositive

full temporal boundaries, model settings, baseline definitions, artifact specifications, and integrity hashes for the contextual data.

4.3.6. Model capacity and computational evidence

Table 4 reports the number of trainable scalar parameters in each neural model, including biases and the ten-site embedding. The counts are calculated from the implemented module dimensions. TimesNet has a task-dependent direct temporal projection, so its parameter count grows with both context and output length. Dilated recurrence changes the temporal connections without adding separate weights for each skipped position. XGBoost uses 150 trees with a maximum depth of three; its capacity is described by those settings.

Training and inference times were not measured consistently across the completed runs. The computational comparison therefore reports parameter counts and the common one-thread policy; runtime comparisons remain for future evaluation.

4.4. Evaluation Metrics

Let y_i be a reference GHI value, $\widehat{y_i}$ its post-processed forecast, $e_i = y_i - \widehat{y_i}$, $\bar{y}$ the mean reference value in the same evaluation vector, and n its number of origin–lead forecast pairs. A target timestamp can occur more than once when forecasts are issued from different origins. Metrics are first calculated separately by site, model estimate, partition, horizon, and horizon scope.

4.4.1. Mean absolute error

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|e_i|\,. \tag{11}$$

MAE is the primary error measure. Expressed in W/m^2, it directly describes the average magnitude of forecast errors. Its linear penalty gives large deviations less influence than a squared-error score [25]. Lower values indicate better performance.

4.4.2. Root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}e_i^2}. \tag{12}$$

RMSE is also expressed in W/m^2, but squaring gives larger deviations more weight. Reporting it alongside MAE reveals whether occasional large errors change the model ordering, a standard concern in deterministic solar-forecast verification [24]. Lower values are better.

4.4.3. Normalized root mean squared error

$$\text{nRMSE} = \frac{\text{RMSE}}{\bar{y}}, \qquad \text{nRMSE}_{\%} = 100\,\text{nRMSE}. \tag{13}$$

The saved data store nRMSE as a dimensionless ratio; percentage values in tables multiply it by 100. Normalization by mean irradiance supports comparisons among sites [26], although a small denominator makes the ratio unstable. nRMSE is left undefined when $\bar{y}$ is numerically zero. Lower values indicate better performance.

4.4.4. Coefficient of determination

$$R^2 = 1 - \frac{\sum_{i=1}^{n}e_i^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}. \tag{14}$$

R^2 measures variation tracked relative to a constant predictor built from the evaluation mean, as defined in Eq. (14). It is undefined for fewer than two targets or a numerically zero reference sum of squares and can be negative when the forecast is worse than the evaluation-mean predictor. Higher values are better. This measure complements the errors expressed in physical units.

4.4.5. Horizon scopes and spatial aggregation

For a declared horizon H, the cumulative scope concatenates leads $1, \ldots, H$ over all valid origins, whereas the exact-lead scope retains only lead H. The two scopes are identical at $H = 1$. This distinction is applied at 24, 48, and 72 hours; 1, 3, 7, 14, and 30 days; and, for the exploratory multi-month task, 1, 3, and 6 months. A horizon label therefore always denotes both a numerical lead and an explicitly named scope. The main rankings use the cumulative scope.

Hourly origins are fixed at local midnight, so exact leads 24, 48, and 72 all select the 23:00 interval on their respective target days. In this test set these nighttime targets and post-processed forecasts are zero: exact-lead MAE and RMSE are zero for every model, while nRMSE and R^2 are undefined. They are retained as audit outputs and do not measure daytime forecast skill.

Within each site, cumulative metrics weight origin–lead pairs equally, including repeated forecasts for a shared target timestamp. They therefore evaluate the issued forecast trajectories

Table 4: Trainable scalar parameters of the neural models under each direct forecasting task. Counts include biases and the ten-site embedding; fixed buffers are excluded.

Task	Context–output	LSTM	TimesNet	DilatedRNN
Hourly	336–72	7,512	27,009	3,096
Daily	365–30	5,790	13,725	2,382
Monthly primary	12–1	4,601	2,758	1,889
Monthly exploratory	12–6	4,806	2,823	1,974

rather than one unique forecast per calendar time. Target periods inside an overlapping window can receive more contributions than periods near the test boundaries.

Let $\mathcal{L}_M$ be the set of locations with a finite site-level value of metric M, and let $L_M = |\mathcal{L}_M|$. The macro result is

$$M_{\text{macro}} = \frac{1}{L_M} \sum_{\ell \in \mathcal{L}_M} M_\ell, \qquad L_M > 0. \tag{15}$$

Undefined site-level values are excluded from the corresponding mean; if none are finite, the macro value remains undefined. All ten sites contribute to the main cumulative results. Equal site weighting prevents a site with more valid forecast pairs from dominating the geographic summary. Macro-MAE is the predeclared primary metric at every resolution, with RMSE, nRMSE, and R^2 reported as complementary measures.

For daily and monthly learned models, seed-level metrics describe variability across five fitted instances on the same test data. Their standard deviation measures sensitivity to the registered seeds; it is neither a forecast prediction interval nor a confidence interval for generalization error. The TimesNet–DilatedRNN paired comparison uses the post-processed five-seed ensemble forecast from Eq. (10). The hourly extension uses seed 42 for both models. For each site, metric, horizon, and scope, the paired difference is

$$\Delta_{\ell,M} = M_{\ell,\text{TimesNet}} - M_{\ell,\text{DilatedRNN}}. \tag{16}$$

A negative $\Delta_{\ell,M}$ is a TimesNet win for the three error metrics; a positive value is a win for R^2. Mean and median differences and the number of site wins are descriptive. Overlapping multi-step windows are not treated as independent observations for an i.i.d. significance test.

5. Results and Discussion

5.1. Primary architecture comparison

Table 5 identifies the lower-MAE architecture at every cumulative checkpoint. All values are test macro-MAEs with equal weight per site, computed from the same origins and targets within each task. Daily and monthly values evaluate the five-seed forecast ensemble; hourly values evaluate seed 42. A cumulative horizon includes all leads up to that checkpoint, rather than only the final lead.

At the terminal horizons, TimesNet led the 72-hour task, with 52.48 versus 52.96 W/m^2 for DilatedRNN. DilatedRNN led the 30-day task (43.85 versus 44.08 W/m^2), the primary one-month task (13.01 versus 13.89 W/m^2), and the exploratory six-month task (14.37 versus 14.46 W/m^2). The largest relative terminal reduction was DilatedRNN's one-month gain: 6.38% relative to TimesNet, calculated from unrounded MAEs. These task-specific outcomes cannot be combined into an overall score because the targets, forecast origins, and aggregation scales differ across tasks.

Table 6 complements MAE with RMSE, nRMSE, and R^2. The terminal RMSE ordering agrees with the MAE ordering: at 72 hours, TimesNet obtained 101.17 W/m^2 versus 104.93 W/m^2 for DilatedRNN; DilatedRNN had the lower RMSE in the daily and monthly tasks. The comparison with the other evaluated models is reported in the Supplementary Material and discussed in Section 5.4.

5.2. Site-paired differences and horizon dependence

Table 7 gives the site-paired comparison; a negative difference denotes lower TimesNet MAE. TimesNet led at 24 and 72 hours by 1.35 and 0.48 W/m^2, respectively; DilatedRNN led at 48 hours by 0.02 W/m^2. Differences are calculated before rounding individual MAEs. TimesNet won seven sites at 24 hours and four at both 48 and 72 hours. Its 72-hour macro advantage therefore reflects the magnitude of local gains, even though DilatedRNN won more sites.

In the daily task, TimesNet led by 0.07–0.17 W/m^2 through seven days and by 0.02 W/m^2 at 14 days. At 30 days, DilatedRNN led by 0.23 W/m^2 and won six sites. DilatedRNN also led the primary one-month task by 0.89 W/m^2 and won six sites. In the exploratory six-month task, TimesNet led the one-month prefix by 0.03 W/m^2 despite winning only four sites. DilatedRNN led by 0.16 and 0.09 W/m^2 at three and six months, respectively, and won seven sites at the terminal horizon. At three months, TimesNet won six sites despite its higher macro-MAE. These reversals show why both the magnitude and the distribution of site-level errors matter.

Figure 3 shows the terminal contrasts at all ten sites. The mixed signs identify locations where the task-level winner has higher MAE, highlighting the geographic variation behind the macro ranking.

5.3. Variability across seeds

Daily and monthly learned models use five random seeds. The 30-day seed-level macro-MAE was 44.29 ± 0.29 W/m^2 for TimesNet and 44.10 ± 0.60 W/m^2 for DilatedRNN. At one month it was 14.62 ± 0.74 versus 13.45 ± 0.35 W/m^2, respectively; at six months, 15.24 ± 0.76 versus 14.77 ± 0.45 W/m^2. Figure 4 reports the means and sample standard deviations of the MAEs computed for individual seed runs. These means

Table 5: Focused architecture comparison: cumulative test macro-MAE in W/m^2. Bold marks the lower unrounded MAE within each row. Δ = $MAE_{TimesNet} - MAE_{DilatedRNN}$ is computed before rounding. Hourly estimates use seed 42; daily and monthly estimates use five-seed ensembles. The complete reference-model benchmark is retained in Supplementary Table S1.

Task	Cumulative horizon	TimesNet	DilatedRNN	Δ	Lower MAE
Hourly	24 h	**49.39**	50.73	-1.35	TimesNet
Hourly	48 h	52.16	**52.15**	+0.02	DilatedRNN
Hourly	72 h	**52.48**	52.96	-0.48	TimesNet
Daily	1 d	**43.20**	43.27	-0.07	TimesNet
Daily	3 d	**43.63**	43.80	-0.17	TimesNet
Daily	7 d	**43.95**	44.08	-0.13	TimesNet
Daily	14 d	**44.09**	44.11	-0.02	TimesNet
Daily	30 d	44.08	**43.85**	+0.23	DilatedRNN
Monthly 12–1	1 mo	13.89	**13.01**	+0.89	DilatedRNN
Monthly 12–6	1 mo	**14.86**	14.89	-0.03	TimesNet
Monthly 12–6	3 mo	14.49	**14.33**	+0.16	DilatedRNN
Monthly 12–6	6 mo	14.46	**14.37**	+0.09	DilatedRNN

Table 6: Terminal cumulative metrics for TimesNet and DilatedRNN, averaged equally across sites. Daily and monthly results evaluate the five-seed ensemble; hourly results use seed 42. MAE and RMSE are in W/m^2; nRMSE is reported as a percentage.

Task	Horizon	Model	MAE	RMSE	nRMSE (%)	R^2
Hourly 336–72	72 h	TimesNet	52.48	101.17	50.41	0.87
Hourly 336–72	72 h	DilatedRNN	52.96	104.93	52.31	0.86
Daily 365–30	30 d	TimesNet	44.08	55.16	26.55	0.53
Daily 365–30	30 d	DilatedRNN	43.85	54.79	26.38	0.54
Monthly 12–1	1 mo	TimesNet	13.89	17.49	8.51	0.88
Monthly 12–1	1 mo	DilatedRNN	13.01	16.02	7.86	0.91
Monthly 12–6	6 mo	TimesNet	14.46	18.24	7.77	0.85
Monthly 12–6	6 mo	DilatedRNN	14.37	17.47	7.47	0.85

need not equal the ensemble MAEs in Table 5, which evaluate the post-processed forecast defined in Eq. (10): the five forecasts are averaged in physical units before the zero floor is applied. The hourly estimate uses one seed and cannot quantify seed variability.

The 0.23 W/m^2 terminal daily ensemble gap is smaller than either architecture's seed-level standard deviation; the six-month gap is only 0.09 W/m^2. In comparisons of runs using the same seed, DilatedRNN had lower terminal MAE in three of five daily runs and four of five runs in each monthly task. Its monthly advantages were therefore more consistent across the evaluated seeds than its terminal daily advantage. Seed standard deviations measure training variability on the same test year; they are neither confidence intervals nor estimates of uncertainty across future years. Comparing an ensemble gap with these deviations is descriptive, not a significance test.

5.4. Reference benchmark and relation to the source protocols

The full benchmark places the two-architecture comparison in context. Supplementary Table S1 and Figure S1 report every evaluated reference model and cumulative checkpoint. XGBoost had the lowest macro-MAE at all hourly checkpoints (50.50 W/m^2 at 72 hours) and at 1, 3, and 14 days. LSTM led at 7 days (43.69 W/m^2), and DilatedRNN led at 30 days (43.85 W/m^2), ahead of XGBoost (43.93 W/m^2). Calendar climatology led both monthly tasks, with 12.57 W/m^2 at one month and 13.03 W/m^2 over six months. Hourly climatology was not evaluated; its missing entries are not zero errors. Consequently, a win over the other focal architecture must not be read as a win over the entire benchmark.

The source studies motivate the hourly TimesNet and monthly DilatedRNN tasks, but their original error values were obtained under different protocols. Here, architecture rankings use predictions with matched inputs, origins, post-processing, aggregation, and metrics within each task.

5.5. Favorable and unfavorable cases

The global case-selection rule identifies the maximum and minimum, over all available sites, origins, and declared cumulative horizons, of $MAE_{DilatedRNN} - MAE_{TimesNet}$. Figure 5 plots both selected forecasts. Weather data were added after the error extremes had been selected. The largest TimesNet gain occurred at GM Factory Zero for the 24-hour forecast beginning on 31 March 2024 in fixed local time: TimesNet and DilatedRNN obtained 45.43 and 109.90 W/m^2, respectively, a 64.47 W/m^2 advantage. NASA POWER provided post-hoc context of 99.64% mean cloud amount and 0.80 mm daily precipitation.

Table 7: Descriptive site-paired cumulative comparison. Δ = $MAE_{TimesNet} - MAE_{DilatedRNN}$ in W/m^2; negative values favor TimesNet (TN), positive values favor DilatedRNN (DR). Wins count sites, not forecast origins. Overlapping multi-step targets are not independent observations.

Task	Horizon	Sites	Mean Δ	Median Δ	TN wins	DR wins	Ties
Hourly 336–72	24 h	10	-1.35	-1.33	7	3	0
Hourly 336–72	48 h	10	0.02	0.62	4	6	0
Hourly 336–72	72 h	10	-0.48	0.50	4	6	0
Daily 365–30	1 d	10	-0.07	-0.04	5	5	0
Daily 365–30	3 d	10	-0.17	-0.15	5	5	0
Daily 365–30	7 d	10	-0.13	-0.23	6	4	0
Daily 365–30	14 d	10	-0.02	-0.09	6	4	0
Daily 365–30	30 d	10	0.23	0.09	4	6	0
Monthly 12–1	1 mo	10	0.89	0.35	4	6	0
Monthly 12–6	1 mo	10	-0.03	0.38	4	6	0
Monthly 12–6	3 mo	10	0.16	-0.08	6	4	0
Monthly 12–6	6 mo	10	0.09	0.50	3	7	0

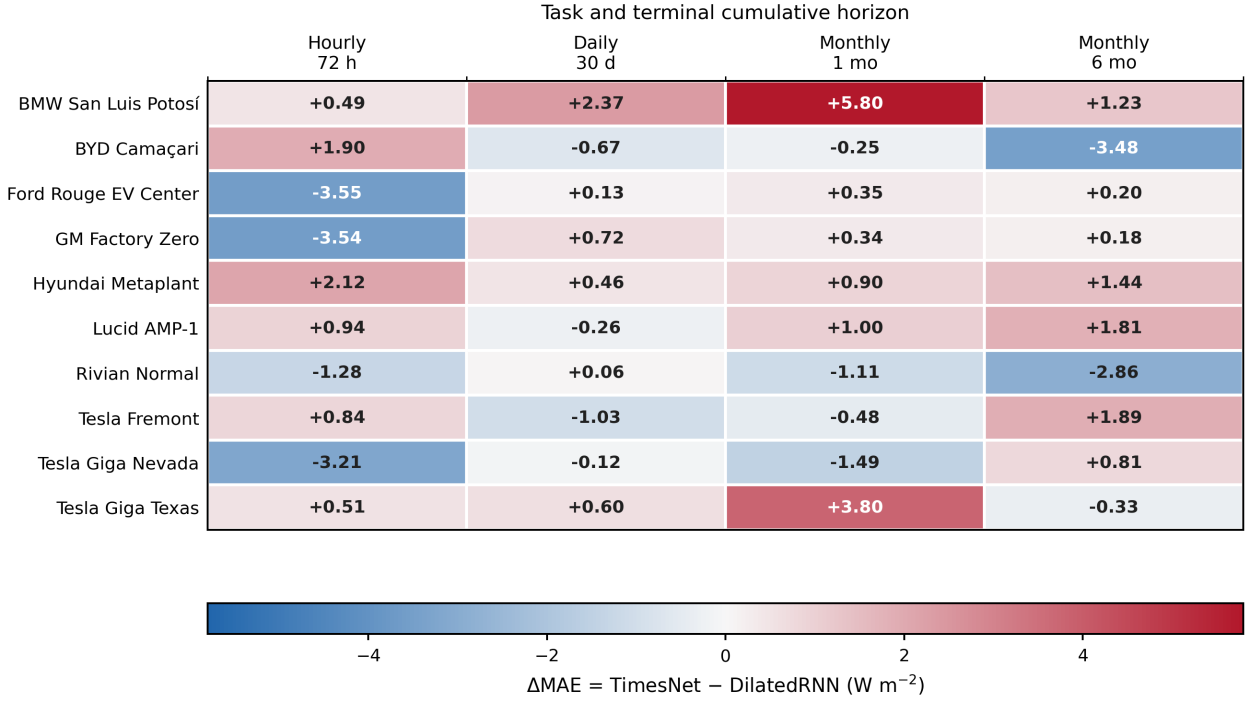


Figure 3: Site-level terminal cumulative contrast $\Delta MAE = MAE_{TimesNet} - MAE_{DilatedRNN}$ in W/m^2. Negative (blue) cells favor TimesNet; positive (red) cells favor DilatedRNN. Columns correspond to 72 h, 30 d, 1 month, and 6 months. Comparisons are descriptive; overlapping multi-step target windows are not treated as independent observations.

The largest deficit occurred at Tesla Gigafactory Texas for the 24-hour forecast beginning on 28 March 2024: 150.02 versus 73.92 W/m^2, a 76.10 W/m^2 TimesNet disadvantage. NASA POWER reported 0.06 mm daily precipitation and 2.14% cloud amount; neither adverse-day threshold was exceeded. An absent flag does not establish the absence of subdaily weather variability. Both cases use the single hourly seed. Selection based on raw MAE contrasts across resolutions can favor tasks with larger absolute error scales, so these extremes cannot establish a cross-resolution ranking.

5.6. Geographic and meteorological context

Figure 1 locates the ten sites within their Köppen–Geiger climate zones. Sites within a country can occupy distinct regimes, while nearby sites may share correlated weather. With only ten purposively selected locations, differences between zones cannot establish a causal climatic effect.

To complement this geographic context, the meteorological selection rule identified an event at Hyundai Metaplant Georgia on 5 August 2024, with 82.66 mm/day corrected precipitation and 99.47% cloud amount. The event was selected independently of model errors. Its weather record is reported in the Supplementary Material, and the following subsection exam-

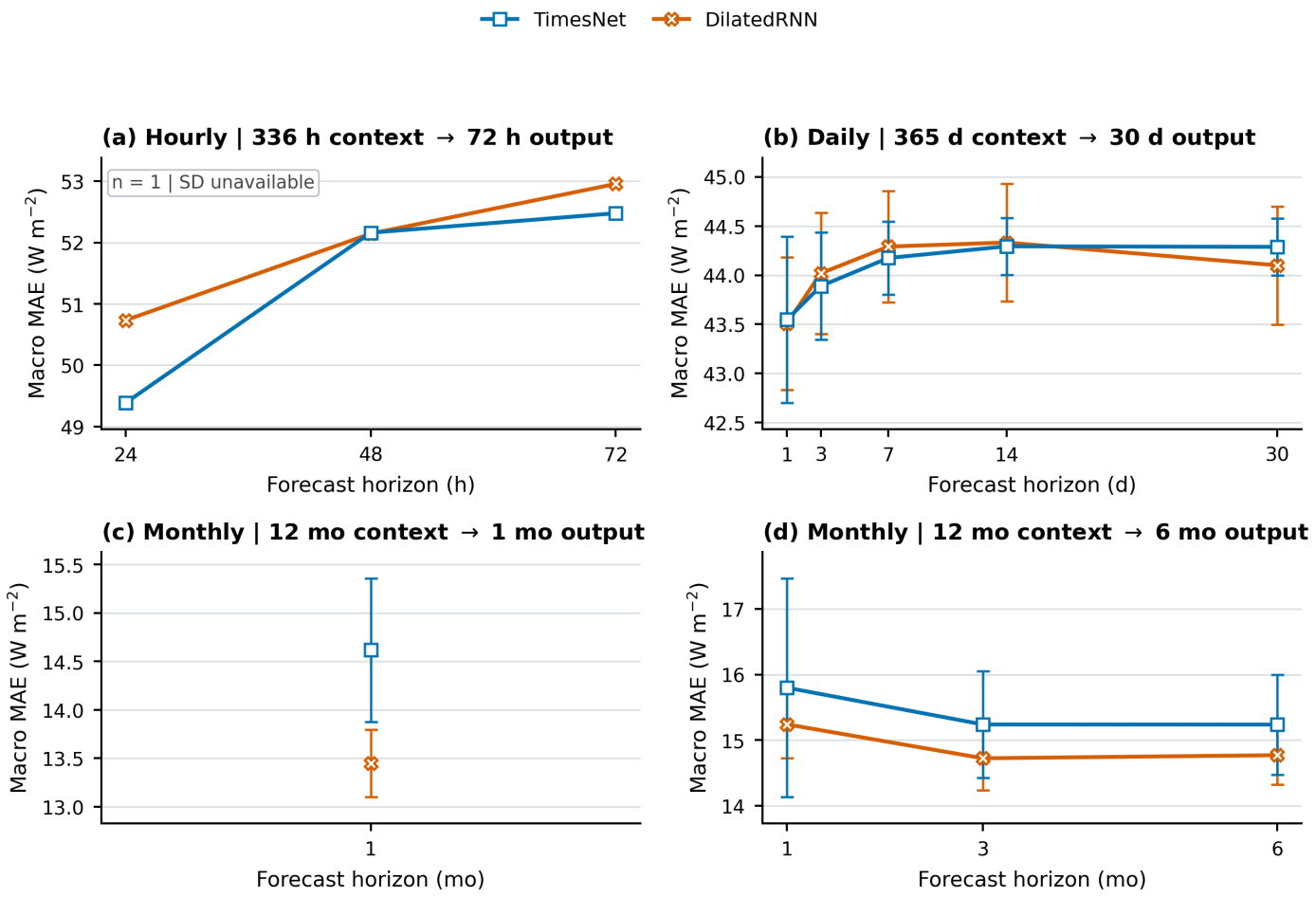


Figure 4: TimesNet and DilatedRNN seed-level macro-MAE: means with error bars spanning one sample standard deviation. Color and marker shape identify each architecture. The hourly extension is shown as $n = 1$ without an error bar. The full learned-model comparison is reported in the Supplementary Material.

ines the forecast profiles around this event and the two error-selected cases.

5.7. *Multi-day irradiance profiles*

Figure 6 extends the two error-selected cases and the independent weather case to seven calendar days each: the event date, three preceding days, and three following days. This fixed extension rule does not optimize the displayed week using model errors. Each panel contains 168 unique hourly targets and seven successive midnight forecast origins. Only leads 1–24 of each 72-hour model output are retained; the context advances daily while the fitted parameters remain fixed. The figure therefore shows seven daily forecasts, not one 168-hour forecast.

Across these seven-day displays, TimesNet and DilatedRNN respectively obtained MAEs of 99.73 and 115.07 W/m^2 at GM Factory Zero, 91.90 and 89.19 W/m^2 at Tesla Gigafactory Texas, and 109.42 and 122.98 W/m^2 at Hyundai Metaplant Georgia. These local window errors are distinct from the single-event errors and the annual macro-MAEs. The additional days reveal underprediction of some high-GHI peaks and overprediction of low-GHI daytime periods by both architectures. At the independently selected Georgia event, both forecasts exceed much of the reference daytime profile. These trajectories show the difficulty both GHI-only models face in following the observed daytime variation.

5.8. *Threats to validity and limitations*

5.8.1. *Construct and data validity*

NSRDB GHI is a satellite-informed physical-model estimate [20], not an on-site pyranometer record. Factory coordinates select product grid cells, so neither the target nor its error quantifies the irradiance at a particular roof surface. The purposive set of ten factory-associated locations is also not a sample of factory electricity demand, photovoltaic output, roof area, or operational constraints. The industrial framing therefore identifies geographically relevant forecasting sites; it does not establish plant-level energy savings.

The matched input contains only GHI available at the forecast origin and site identity. This restriction controls predictor availability among learned models, but excludes potentially useful weather information, such as cloud motion, numerical weather predictions, and forecasts of aerosols and temperature. A site embedding can also encode persistent location-specific behavior and does not demonstrate transfer to an unseen factory. Fixed local offsets produce regular hourly grids but do

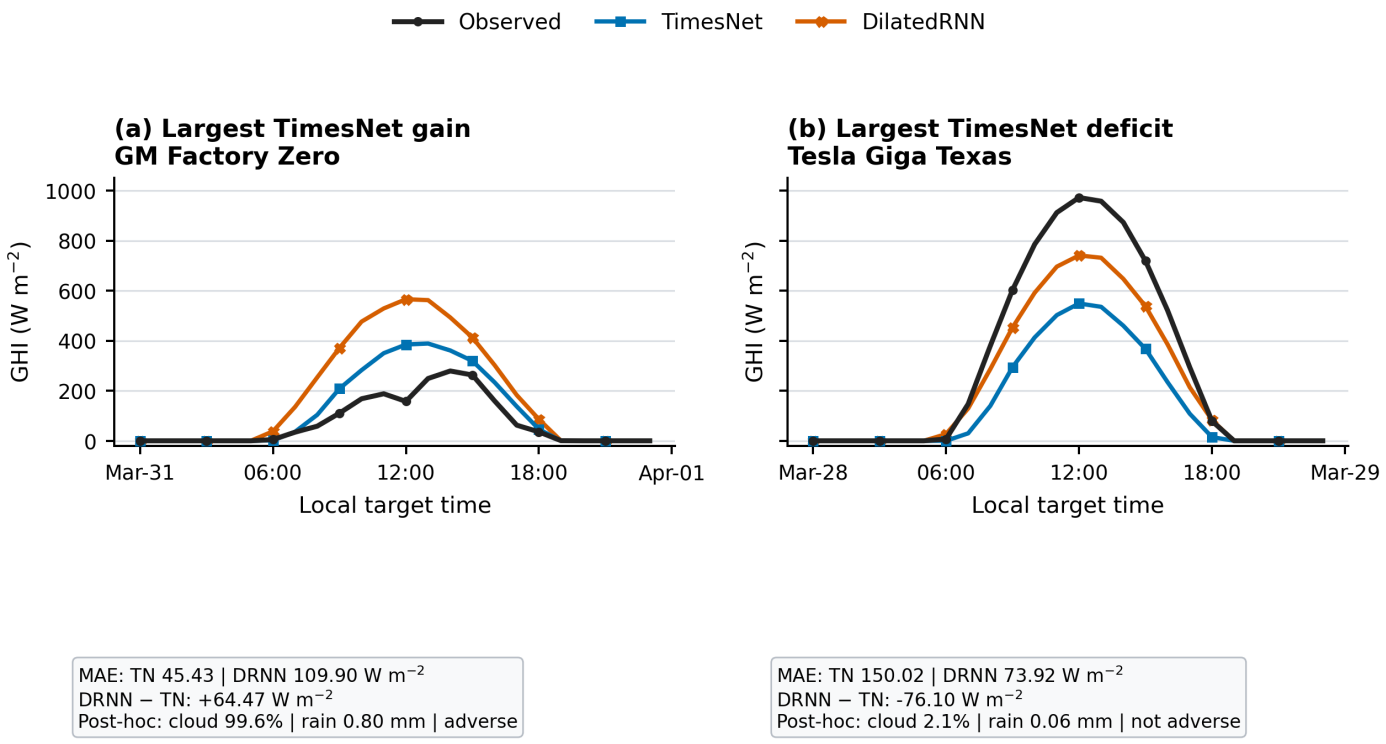


Figure 5: Reference GHI and forecasts for the globally selected largest TimesNet gain and deficit. Panels share a GHI scale. MAEs and descriptive weather records refer to the selected 24-hour windows; weather data were added only after case selection.

not reproduce daylight-saving clock changes. Moreover, the daily/monthly archive was retrieved independently of the hourly archive. The comparison across resolutions therefore does not use controlled reaggregation of a single set of native records.

5.8.2. Temporal and statistical validity

All test evidence is retrospective and restricted to 2024. It does not measure prospective robustness under later satellite-product revisions, climate nonstationarity, or new sites. The retrospective information set assumes that each completed historical aggregate is immediately available; product publication delays and operational data latency are not simulated. The primary monthly task has only 48 pre-test refit origins per site, and the exploratory six-month task has seven test origins per site. These small samples limit the evidence available for fitting and evaluating monthly models relative to the hourly and daily tasks.

Targets from adjacent multi-step origins overlap, creating strongly dependent errors. Site-first aggregation avoids allowing that overlap to change spatial weights, but it does not make the origins independent. The paired summaries therefore use ten locations as descriptive units, without applying an i.i.d. significance test to the overlapping target points. Nearby sites may also share weather and model errors, so their summaries cannot be assumed to be independent statistical replicates.

5.8.3. Model-comparison validity

TimesNet and DilatedRNN share inputs, origins, seeds, optimization limits, and post-processing in the daily and monthly tasks, but their parameterizations and inductive biases are not identical. The frozen configuration is a controlled comparison rather than an exhaustive architecture-specific search. Moreover, the hourly extension evaluates DilatedRNN using seed 42 against previously generated comparator forecasts, whereas daily and monthly estimates use five seeds. Comparisons across resolutions therefore reflect these protocol differences, and seed variability cannot be assessed for the hourly task. The reported TimesNet is a compact variant with sample-specific Fourier-period selection; the findings do not establish a ranking of every implementation bearing either architecture name.

Neural epoch selection minimizes normalized validation MSE, whereas the main test ranking minimizes macro-MAE in W/m^2. These objectives weight errors differently, and equal selection rules do not establish each model's best achievable MAE. Calendar baselines also use explicit target calendar labels and historical lookups unavailable as features to the learned models. Their performance is a useful operational reference, but does not isolate architecture under identical feature representations.

The neural parameter budgets also differ by task, as reported in Table 4. Equal optimization rules do not imply equal representational capacity, and the comparison cannot isolate whether a result comes from recurrence, periodic convolution, parameter count, or their interaction. Training and inference runtimes were not measured consistently across runs, so the results do not support a hardware-efficiency ranking.

The zero floor enforces nonnegative predictions at all resolutions. The additional hourly nighttime mask uses deterministic

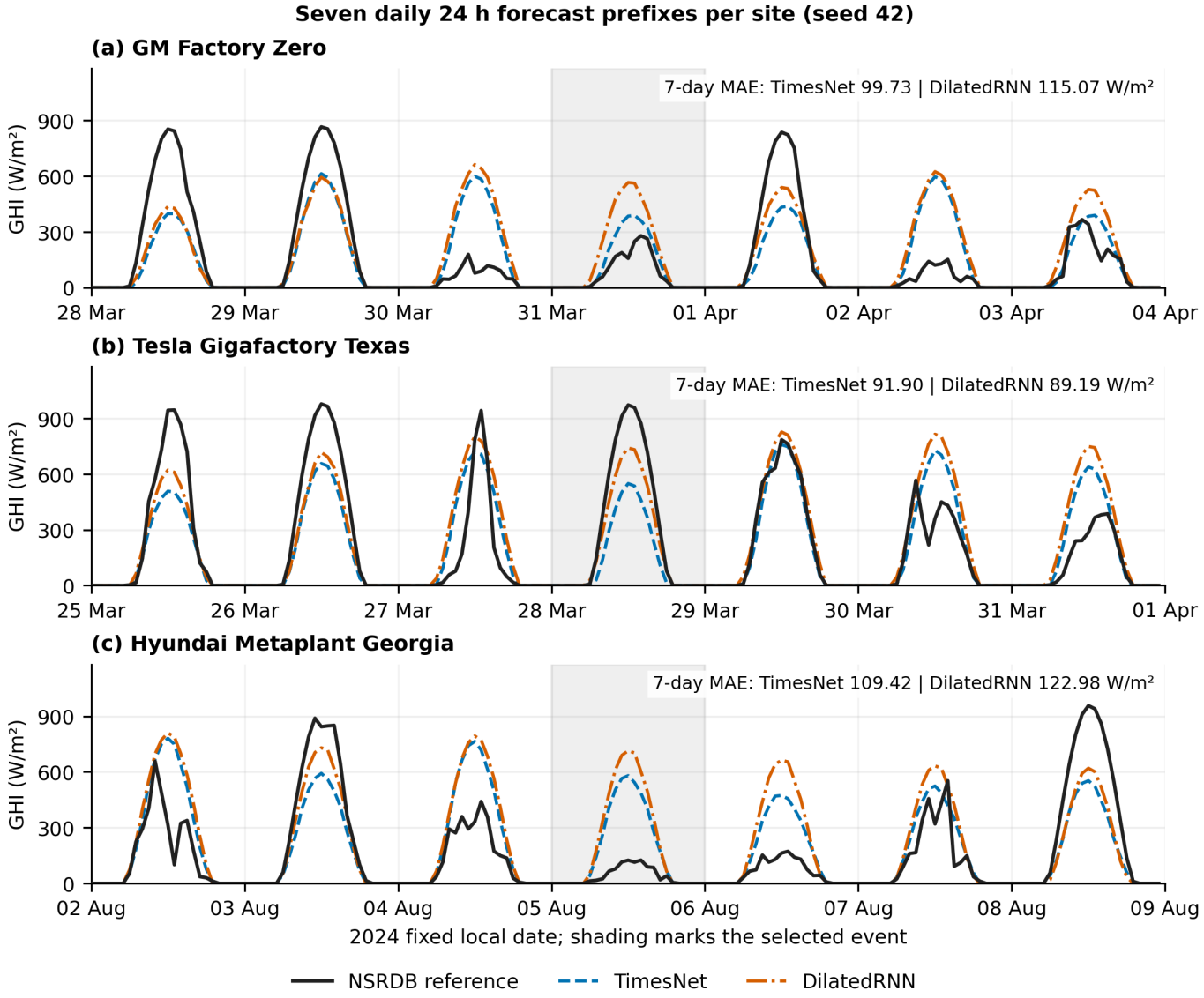


Figure 6: Seven-day hourly GHI profiles around the two error-selected cases and the error-independent NASA POWER event. Each curve joins seven 24-hour forecast prefixes issued on successive local midnights; it is not a single seven-day forecast. Shading marks the original selected day. MAEs use all 168 displayed targets per site, including nighttime; all forecasts use seed 42 and the common hourly post-processing. Dates follow the fixed local-time index, and all panels share a GHI scale.

solar geometry, but no common upper physical envelope is imposed. Consequently, implausibly high daytime extrapolations remain reflected in the reported errors.

Hourly metrics retain all target hours after the common nighttime mask. Exact nighttime zeros are part of the operational 72-hour trajectory, but they also dilute the contribution of daylight errors to the all-hour average. The reported hourly values must therefore be interpreted as full-trajectory metrics. In particular, exact leads 24, 48, and 72 sample only the nighttime 23:00 intervals and cannot distinguish model skill in this experiment. Daylight-only evaluation and complete lead profiles are needed to assess the irradiance-producing hours separately.

5.8.4. Contextual-data validity

The Köppen–Geiger labels are climate classes extracted from cells of the present nominal 1-km map [46]; they do not resolve microclimates within a cell or conditions on a specific roof. Class boundaries are uncertain, as made explicit by the 50% confidence of the BMW San Luis Potosí assignment. Climate labels are descriptive strata, not model inputs and not explanations of individual forecast errors.

NASA POWER is likewise a gridded, model-based external product rather than local rain-gauge or ceilometer evidence [48]. The 1 mm/day precipitation and 80% cloud thresholds provide reproducible post-hoc tags, but thresholding cannot establish that weather caused a particular error. These weather observations provide context for the selected cases, while model rank-

ings are based on the full aggregate evaluation.

6. Conclusion

The matched comparison shows that the relative performance of TimesNet and DilatedRNN depends on the GHI forecasting task. Across ten factory-associated locations, inputs available at the forecast origin, data partitions, forecast origins, post-processing, metrics, and location weights were controlled within each task. The results describe forecasts of the NSRDB solar resource, without quantifying factory demand or photovoltaic output.

For cumulative macro-MAE, TimesNet led at 24 and 72 hours and at the daily checkpoints up to 14 days. DilatedRNN led at 48 hours, 30 days, and in the primary one-month task. In the exploratory six-month task, TimesNet led at the one-month prefix, while DilatedRNN led at the three- and six-month prefixes. Local win counts sometimes disagreed with the macro ranking: DilatedRNN achieved lower terminal MAE at six of the ten sites in each of the hourly, daily, and primary one-month tasks, and at seven sites in the exploratory six-month task.

The complete benchmark puts these architecture-specific advantages in perspective. XGBoost led all hourly checkpoints and the daily checkpoints at 1, 3, and 14 days; LSTM led at seven days, and DilatedRNN led at 30 days with 43.85 W/m^2, compared with 43.93 W/m^2 for XGBoost. Calendar climatology led the monthly checkpoints. Neither TimesNet nor DilatedRNN was consistently best across tasks.

Daily and monthly estimates use five-seed ensembles, and several differences are small relative to the observed seed variability. The single hourly seed, overlapping targets, one test year, and unequal parameter budgets limit generalization. Post-hoc weather cases provide context but do not explain model differences causally.

Future evaluation should include prospective years and unseen sites, multiple hourly seeds, daylight-only metrics, measured computational cost, and weather forecasts available at issuance. The matched protocol and origin-level predictions provide a basis for testing these extensions. Together, the task-level rankings, site-level differences, and reference benchmarks show why architecture choice must be evaluated for the intended forecasting task.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data and Code Availability

Source code and protocol definitions are maintained at `https://github.com/enggusttavomn/TCC`. The comparisons reported here use the corrected multi-resolution archive identified in the Supplementary Material. That full forecast bundle is not included in the current public repository snapshot; access to it is required to reproduce the reported metrics from individual predictions. External NSRDB, Köppen–Geiger, and NASA POWER products remain subject to their respective providers' access and licensing conditions.

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